

Massive transfusion in the pediatric population: A systematic review and summary of best-evidence practice strategies

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BACKGROUND:	Pediatric patients require massive transfusion (MT) in a variety of settings. Multiple studies of adult MT support balanced ratio transfusion to improve outcomes; however, it is unclear if these findings can be extrapolated to pediatric populations. The use of balanced transfusion ratios, MT protocols, hemostatic adjuncts, and even the definition of a MT in children are all open questions. This review presents details of care from current practices in pediatric MT and summarizes practice strategies while providing insight from our single-center experience.
METHODS:	PubMed, EMBASE, and Web of Science were searched using MeSH index and free-text terms for articles from 1946 to 2017. Articles were independently reviewed by two reviewers. Studies were assessed for definition of MT, factors predicting MT, MT complications, blood product ratios, hemostatic adjuncts, protocol logistics, and clinical outcomes.
RESULTS:	A heterogeneous composite of 29 articles was included in the analysis. Of these, 45% reported a formal transfusion protocol or adopted one during the study. Seven unique definitions of pediatric MT were reported; the most common was >1 total blood volume within 24 hours. A total of 18,369 patients were assessed, and 1,163 received MT (6.3%). Overall mortality for patients requiring MT in studies reporting mortality was high (range 14.7% to 51.2%). We identified 14 patients receiving MT at our center with an age range of 8 months to 18 years and average transfusion of 38.1 mL/kg red blood cells (range: 22.1 mL/kg to 156.7 mL/kg).
CONCLUSIONS:	Current practices of pediatric MT demonstrate a variety of site-specific interventions with a persistently high mortality rate. A national focus on improving techniques of MT in children has the potential to save the lives of these children. (<i>J Trauma Acute Care Surg.</i> 2019;86: 744–754. Copyright © 2019 Wolters Kluwer Health, Inc. All rights reserved.)
LEVEL OF EVIDENCE:	Systematic review, levels IV and V.
KEY WORDS:	Massive transfusion; pediatric surgery; coagulopathy; trauma.

Pediatric patients require massive transfusion (MT) in a variety of settings ranging from trauma to refractory bleeding during surgery. MT in adult populations has been studied extensively, and current evidence suggests that a balanced transfusion approach improves achievement of hemostasis and mortality.^{1,2} However, because of developmental differences in hemostasis and variations in mechanism of injury, extrapolating adult strategies to children may be inappropriate.³ Limited data exist to support optimal MT strategies in pediatric patients and rather demonstrate the wide range of current practices.

The aim of this systematic review was to describe current practices in pediatric MT and provide clinicians with a summary of existing literature regarding the clinical care for pediatric patients requiring MT. In addition to the systematic review, we sought to review the experience at our trauma center after implementing a blood bank–based pediatric massive transfusion protocol (MTP) designed to achieve balanced red blood cell (RBC) to fresh frozen plasma (FFP) ratio deployment, with attention to protocol adherence and patient outcomes. As clinical trials in this area are non-existent, we aimed to provide a foundation to

develop an approach to care in pediatric patients requiring large amounts of blood product.

METHODS

Search Strategy

We searched for articles pertaining to pediatric MT using MeSH terms “pediatrics”, “blood volume”, “massive”, and “transfusion”. We reviewed all available citations in PubMed, EMBASE, and Web of Science from January 1946 to December 2017. Articles were restricted to human subjects and written in English only. A protocol for this systematic review is not included.

Selection Criteria

To be eligible for inclusion, studies needed to be (1) full-text, peer-reviewed articles, and (2) describe an MTP, or (3) include pediatric patients receiving MT, or (4) provide guidelines for pediatric MT. Studies were excluded if they did not contain patients receiving MT, did not document guidelines pertaining to MT, or contained primarily adult patients.

Data Extraction and Quality of Included Studies

All articles derived from the three databases were independently screened by two reviewers. Articles were subject to screening via title, abstract, and ultimately a full-text review. At each stage of screening, articles were compared against the inclusion and exclusion criteria above. Disagreements were resolved by consensus-based discussion. Our search returned a heterogeneous composite of articles, including retrospective

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analyses, case series, case reports, review articles, prospective cohort studies, quality improvement assessments, and surveys.

Data Collection and Analysis

Because of the heterogeneity of studies collected, no meta-analysis was performed. The measures considered for analysis were aspects of clinical management specific to MT. These included definitions of pediatric MT, blood product ratios released and transfused, factors predicting MT, complications of MT, hemostatic adjuncts, protocol logistics, and patient-specific outcomes (mortality, length of stay, ICU length of stay, ventilator days, time to first blood product, thrombosis/coagulopathy). Descriptive statistics were presented, with averages reported as mean values unless otherwise stated and proportions reported in percentages.

Local Experience With Pediatric Massive Transfusion

After approval by the Duke University Institutional Review Board (IRB no. 00080108), we conducted a retrospective review of MTP activations for children ≤ 18 years old from 2015 to 2017 as defined in the Duke University Medical Center blood bank database. Medical and transfusion records were queried for patient demographics and outcomes, MTP activation logistics, and administered blood products. The MTP targeted a 1:1 RBC/FFP ratio with deployment of blood product in successive rounds. Hemostatic adjuncts were not included in the formal protocol and were used only at physician discretion. Likewise, no set laboratory values for monitoring were part of the protocol. Univariate descriptive statistics were used to describe demographics and logistic findings from all MTP activations.

RESULTS

Overview of Articles Meeting Inclusion Criteria

The literature search identified 3,126 publications, which was reduced to 1,941 after removal of duplicates (Fig. 1). After initial reviews by title and abstract for relevance and completeness, 41 articles were identified for full-text review, and 29 were selected for final analysis.^{4–32} The articles chosen for analysis were heterogeneous, consisting of 13 retrospective analyses, four case series, four case reports, four review articles, two prospective cohort studies, one quality improvement study, and one survey of transfusion protocols. This heterogeneity in studies precluded a formal meta-analysis, and a qualitative assessment of the studies was performed. Outcomes and practices identified from the articles are summarized in Tables 1–5. Table 1 provides a summary of all studies included in the analysis.

Defining Pediatric MT

Our analysis considered how various centers may define pediatric MT and how the definition has evolved over time. We found that 24 (82.8%) articles reported a total of seven unique definitions of pediatric MT. Nearly all definitions include a volume threshold and a time interval. The most liberal volume per unit time posited as MT was greater than 50% total blood volume (TBV) replacement within 1 h, whereas the lowest volume per unit time posited as MT was greater than 50% TBV within 24 h.^{21,22,29} The most commonly cited definition was greater than or equal to one TBV transfused within 24 h.^{6,8,9,14–16,18,20,24,26} This definition was popular among articles published before

2015. However, the majority of most current articles since 2015 use the definition of greater than 40 mL/kg total blood product within 24 h.^{4,12,23,25,27,30–32} Additionally, some propose that greater than 50% TBV replacement within 1 h or 3 h constitutes MT.^{13,29}

Ratios of Blood Products Utilized in Pediatric Massive Transfusion Protocols

A total of 15 studies reported details of the MTP used by their institution or consortium. Of those reporting details regarding emergency release products, 55.6% (5/9) reported emergency release blood product as part of the protocol. One reported releasing FFP along with RBC in the emergency release products.¹⁸ Of the studies reporting products released with each round of transfusion, 77.8% (7/9) reported weight-based ratios for pediatric patients. The amount of weight categories varied from two to five. The most common target RBC/FFP ratio was 1:1. However, the actual RBC/FFP ratios delivered varied unpredictably.

Triggers and Goal-Directed Therapy During Protocol Activation

Activation criteria for pediatric MTPs include both subjective and objective triggers (Table 2). We found that the majority of protocols were activated at physician discretion.^{9,15,22,32} This is consistent with survey data from Horst et al. who found that 89% of protocols were activated at physician discretion.¹⁹ Although some protocols simply defined the trigger as physician discretion, others provided objective supporting factors. Hwu et al. used >40 mL/kg total blood product administration as a signal for activation.²² Others defined their threshold more leniently, with the supplementation of any fluid including crystalloid >40 mL/kg as a criterion for activation.^{12,23} Hemorrhagic shock with persistent or ongoing bleeding was also often factored into the clinical decision-making process.

Of the studies reporting protocols, 78% (7/9) listed initial and repeat laboratories used for monitoring the effects of blood product administration. However, only 29% (2/7) included target laboratory values from these monitoring tests to indicate that the transfusion protocol was achieving goal-directed hemostasis. Common initial laboratories used included type and screen, complete blood count, arterial blood gas, prothrombin time (PT), international normalized ratio (INR), partial thromboplastin time (PTT), and calcium. Only 29% (2/7) reported using viscoelastic hemostatic assays (rotational thromboelastometry [ROTEM] or thromboelastography [TEG]) to monitor coagulopathy. Repeat laboratory intervals were typically defined by time rather than additional transfusion volume increments, with typical intervals of 30 min to 1 h. Of the studies defining goal-directed therapy laboratory standards, thresholds included platelet counts $>50,000$ – $100,000$ and fibrinogen >1.0 g/L.^{11,23}

Factors Associated With Requirement of Massive Transfusion

A total of eight articles described patient-specific factors that were associated with receipt of MT. A more severe Injury Severity Score (ISS) was the most commonly cited factor associated with a requirement for MT. Of the six articles citing this value, Shroyer et al. demonstrated a threshold ISS ≥ 25 whereas

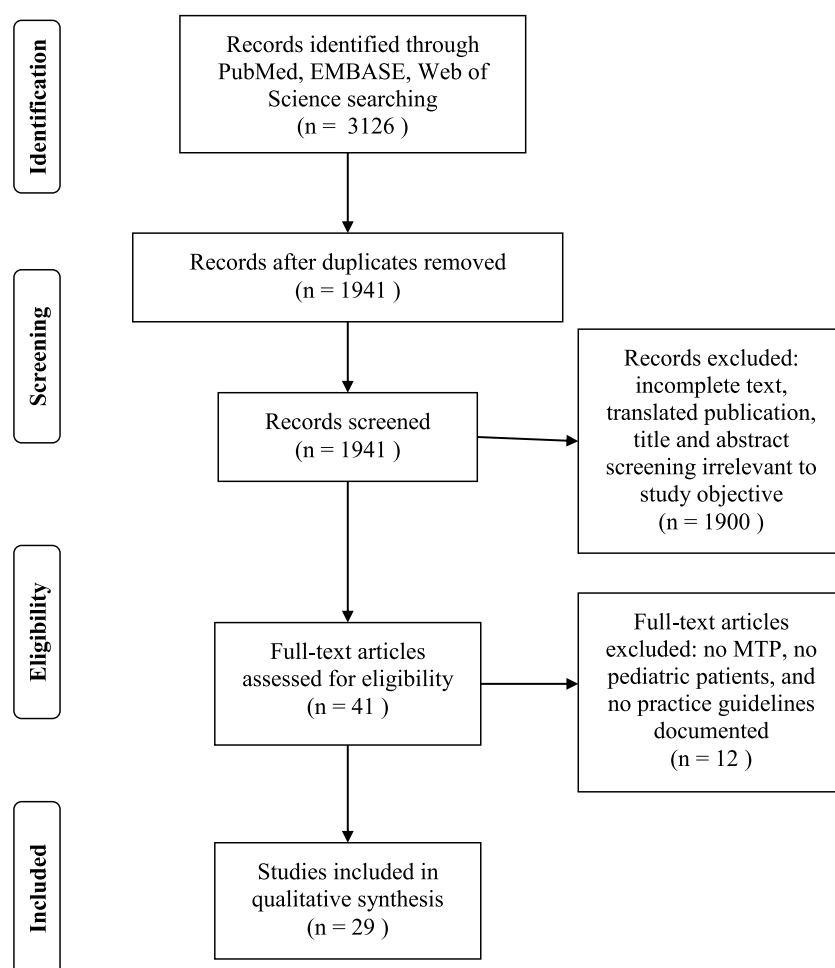


Figure 1. Flowchart of article selection pathway for the systematic review.

a lower threshold was described by Edwards et al.^{17,30} Several measures of coagulopathy were cited, and PTT >32, >36, and elevated INR (>1.5) were all associated with MT.^{9,20–22,27} Although the majority of vital signs at admission were not commonly described, hypothermia (<35°C) was an associated patient factor.^{21,22,27,30} A penetrating mechanism of injury was more commonly cited with MT than blunt trauma; however, two articles describing this mechanism of injury were from military populations, which may be less reflective of injury patterns among civilian populations.^{17,27} Patient age and sex were not reliably associated with MT.^{9,17,20,22,25,27}

Complications of Massive Transfusion

Ten articles described complications presumed secondary to receiving MT (Table 3). The two most commonly cited complications included coagulopathy and hyperkalemia. Coagulopathy in MT arises in part as a result of dilution after receiving large volumes of blood product and crystalloid. Unlike hereditary coagulopathies like hemophilia, this dilutional coagulopathy involves derangements of many coagulation factors and is frequently complicated by other hematologic derangements in the context of trauma or other instigating hemorrhagic insult. This multifactorial problem can be difficult to correct. The

authors offer suggestions to avoid or mitigate this risk of coagulopathy with approaches that include the minimization of crystalloid use and inclusion of platelets and fibrinogen transfusions, two components greatly decreased by dilutional effects. Regarding dilutional thrombocytopenia, Cote et al. suggest prophylactically administering platelets if platelet count is <100,000, as well as vigilance that a child may require platelet transfusion after 1 TBV replacement if they have a baseline platelet count <150,000.¹⁰ Additionally, if abnormal bleeding occurs in the setting of a platelet count >50,000, 0.3 U of platelets/kg body weight is adequate initial therapy.

Hyperkalemia in MT results from the release of potassium into circulation secondary to lysis of large amounts of red blood cells or storage of older cells transfused. Hyperkalemia is particularly concerning because it can lead to asystolic cardiac arrest. To mitigate the rise in serum potassium, suggested strategies involved two major approaches: (1) using blood with less storage lesions and (2) reducing the amount of mechanical stress on RBCs during transfusion. Buntain et al. highlight that transfusing irradiated red cells increases the likelihood of hyperkalemia and recommend that they should be given only when strictly indicated and should be irradiated less than 96 h before transfusion.⁸ Particularly in infants, Lee et al. suggest transfusing RBCs less than

TABLE 1. Summary of Articles on MT Meeting Inclusion Criteria

Article	Study Design	Study Setting	Number receiving MT (Total n)	Age Range in Years (Mean)	Formal MTP	Overall Mortality for Patients Receiving MT (%)
Shroyer 2017	Retrospective review	Multi-institutional (>900) NTDB-participating hospitals	173 (13,523)	≤14 (NA)	No	50.6
Acker 2016	Retrospective review	Multi-institutional (2) pediatric trauma centers (Level I and II)	4 (50)	4–15 (9)	Yes	
Horst 2016	Survey	Multi-institutional (46) children's hospital, units in general hospitals			Yes	
Hwu 2016	Retrospective review	Single-institution ACS Level I pediatric trauma center	43 (235)	<18 (9)	Yes*	51.2
Navaratnam 2016	Case series	Single-institution children's hospital	5 (5)	7–23 (16.2)	No	0
Smith 2016	Retrospective review	Single-institution Level I adult and pediatric trauma center	19 (96)	<18 (9.5)	No	
Sparkle 2016	Case report	Single-institution pediatric surgery center	1 (1)	1 (1)	No	
Edwards 2015	Retrospective review	Multi-institutional US military treatment facilities	77 (907)	0–14 (NA)	No	25
Hwu 2015	Retrospective case-control	Single-institution ACS Level I pediatric trauma center	44 (303)	0–18 (NA)	Yes*	
Neff 2015	Retrospective review	Multi-institutional US combat support hospitals	443 (1,113)	6–14 (10)	No	14.7
Eckert 2014	Retrospective review	Single-institution US military treatment facility	76 (766)	<18 (11)	Yes	
Kua 2014	Review article	Single-institution children's hospital			Yes	
Lee 2014	Case series	Multi-institutional children's hospitals, pediatric surgery centers	9 (9)	0–16 (2.8)	No	33
Livingston 2014	Retrospective review	Single-institution Level I adult and pediatric trauma center	13 (435)	0–17 (10.5)	No	38
Agrawal 2013	Case report	Single-institution pediatric transplant center	1 (1)	10 (10)	No	
Diab 2013	Review article	Single-institution children's hospital			Yes	
Huang 2013	Retrospective review	Single-institution pediatric transplant center	13 (198)	<18 (3.5)	No	
Nosanov 2013	Retrospective review	Single-institution Level I adult and pediatric trauma center	105 (403)	≤18 (12.4)	Yes	18.1
Arul 2012	Prospective cohort	Single-institution British military hospital	6 (82)	0–15 (8)	No	33
Chidester 2012	Prospective cohort	Single-institution Level I pediatric trauma center	22 (55)	0–28 (9.6)	Yes	45
Craig 2012	Quality improvement	Multi-institutional transfusion and trauma committees			Yes*	
Hendrickson 2012	Retrospective review	Single-institution Level II pediatric trauma center	53 (102)	<18 (6.2)	Yes*	38
Dehmer 2010	Review article				Yes	
Dressler 2010	Case report	Single-institution Level I pediatric trauma center	1 (1)	9 (9)	Yes	
Downes 2001	Review article				No	
Buntain 1999	Case report	Single-institution pediatric surgery center	1 (1)	1 (1)	No	
Brown 1990	Retrospective review	Single-institution pediatric surgery center	11 (40)	0–12 (3.1)	No	
Cote 1985	Case series	Multi-institutional pediatric surgery centers	26 (26)	0–17 (5.9)	No	0
Schroeder 1969	Case series	Single-institution pediatric surgery center	17 (17)	0–11 (NA)	No	6

*Formal MTP adopted during the study.

MT, massive transfusion; MTP, massive transfusion protocol.

TABLE 2. Pediatric MT Protocol Triggers and Monitoring

Article	MT Initiation Trigger	MT Monitoring	MT Stop Trigger/Target Results During Ongoing Bleeding
Horst 2016	Of 46 sites, prevalence of MT initiation trigger: Physician discretion: (89%) Vital sign/physical examination: (17%) Laboratory values: (7%) Mechanism of injury: (9%) Blood product volume administered: (15%)	Routine laboratory measures after MTP activation: (50%) INR/PT/PTT (49%) Potassium (24%) Hb/Hct (47%) Base deficit (18%) Fibrinogen (44%) Lactate (18%) PLT (40%) TEG (9%) Ionized calcium (29%) Lab time interval specified: q30min (11%) q6h (2%)	
Hwu 2016	Physician discretion with indicators: >40 mL/kg total blood product administered Clinical concern for acute imminent blood loss		Physician discretion
Sparkle 2016	Physician discretion	Repeat laboratories (q20–30 min): ABG Electrolytes	Physician discretion
Kua 2014	Hemorrhagic shock from uncontrolled bleeding >40 mL/kg of non-blood products or >20 mL/kg of blood products administered	Initial laboratories: CBC ABG PT/PTT Calcium Fibrinogen Repeat laboratories: as above (after each blood package)	Bleeding stabilized PLT $\geq 50 \times 10^9$ (no significant head injury) PLT $\geq 100 \times 10^9$ (significant head injury) pH >7.2 Base deficit >6 Calcium >1.1 mmol/L PT/APTT <1.5 $\times$ normal INR ≤ 1.5 Fibrinogen >1.0 g/L Lactate <3 mmol/L
Diab 2013			Physician discretion
Chidester 2012	Physician discretion ≥ 80 mL/kg RBC within 24 h or ≥ 40 mL/kg within 12 h (actual or anticipated)	Initial laboratories: T&S Hb PLT PT/PTT Fibrinogen Potassium Blood pH Laboratories after first 80 mL/kg RBCs: as above (without T&S) + Fibrin degradation products + Calcium Laboratories after every 40 mL/kg RBCs thereafter: as above + Lactate	
Craig 2012	>20 mL/kg RBC or >40 mL/kg any fluid within preceding 1 h + Ongoing bleeding + Hypovolemic shock and/ or coagulopathy	Initial laboratories: Crossmatch CBC PT/APTT fibrinogen Electrolytes calcium ABG lactate TEG/ROTEM (if available) Repeat laboratories: as above (without crossmatch and ABG)	Hb >10 g/dL PLT $>75 \times 10^9$ /L Fibrinogen >1.0 g/L PT ratio <1.5 APTT ratio <1.5
Hendrickson 2012	Emergency release blood products requested for trauma patient (administered or anticipated)	Initial laboratories: Hb Hct PLT PT/PTT Fibrinogen	
Dehmer 2010	Shock with persistent hemodynamic instability Ongoing bleeding after 40 mL/kg crystalloid infusion	Initial laboratories: calcium blood pH Repeat laboratories: as above	

Continued next page

TABLE 2. (Continued)

Article	MT Initiation Trigger	MT Monitoring	MT Stop Trigger/Target Results During Ongoing Bleeding
Dressler 2010	Physician discretion	Initial laboratories: T&S CBC Blood gas lactate Hb ionized calcium Electrolytes PT/INR Fibrinogen TEG Repeat laboratories (q1h): as above (without T&S)	Physician discretion

MT, massive transfusion; PLT, platelets; TEG, thromboelastography.

7 days old if available, and Downes et al. suggest washing the cells before transfusion if older than 7 days.^{14,24}

Use of Hemostatic Adjuncts in Pediatric MT

Acellular hemostatic adjuncts utilized in addition to RBC, FFP, and platelets included recombinant FVIIa (rFVIIa), tranexamic

acid (TXA), fibrinogen concentrate, prothrombin complex concentrates, and cryoprecipitate (Table 4). No specific dosing guidelines were suggested for fibrinogen concentrate or prothrombin complex concentrates. Cryoprecipitate was the most commonly cited adjunct used in the reported studies. Reported dosing was variable with some including weight-based units of

TABLE 3. Complications Associated With MT in Pediatric Patients

Complication	Key Aggregate Recommendations	Article
Coagulopathy	1. Avoid overuse of crystalloids 2. Baseline PLT <150 may require PLT transfusion after 1 TBV loss 3. Consider prophylactic PLT transfusion with PLT <100 4. 0.3 U of PLT/kg is adequate initial therapy for abnormal bleeding with PLT count >50 5. 1 U cryoprecipitate/4 kg increases fibrinogen level by 25–50 mg/dL 6. For DIC, administer 2–4 U FFP for every 10 U of RBC	Livingston 2014 Hendrickson 2012 Dressler 2010 Downes 2001 Cote 1985 Schroeder 1969
Hyperkalemia (with cardiac arrest)*	1. Use larger bore (>23 gauge) peripheral IV catheters preferentially over central venous lines for MT 2. Avoid using rapid infusers if possible 3. Use stored RBCs <1 week old, particularly in infants 4. Wash RBCs if stored >1 week 5. Reduce the plasma volume of RBC units 6. Transfuse RBCs within 96 h of irradiation	Lee 2014* Agrawal 2013* Downes 2001 Buntain 1999* Brown 1990 Schroeder 1969
Hypothermia	1. Use fluid warmer to warm RBC to body temperature (37°C) 2. Use Bair hugger, heating lamps, and warmers for inhaled gases	Dressler 2010 Downes 2001 Schroeder 1969
Acidosis	1. If pH < 7.2: administer bicarbonate or tris-hydroxymethyl aminomethane (THAM) 2. Acid load buffer: sodium bicarbonate 1.0 mequiv/kg/(25% EBV loss)	Dressler 2010 Schroeder 1969
Hypocalcemia	1. Administer calcium chloride 20 mg/kg for each abnormal laboratory value recorded 2. Administer 1 mL calcium gluconate 10% IV for the first 100 mL of blood loss, and 0.75 mL for subsequent 100 mL losses 3. Prophylaxis with calcium chloride 1.5 mg/kg/min IV with cardiac monitoring (if unresponsive, consider magnesium chloride)	Dressler 2010 Downes 2001 Schroeder 1969
Rh– incompatibility	1. Red cell exchange with Rh– units within days of transfusion (target: Rh+ circulating cells to 10%) then give anti-D immunoglobulin (10–15 µg/mL red cells) 2. If smaller amount of Rh+ cells transfused, administer anti-D immunoglobulin (10–15 mg/mL red cells)	Downes 2001

MT, massive transfusion; TBV, total blood volume; PLT, platelets; RBC, red blood cells.

TABLE 4. Hemostatic Adjuncts Used in MT

Hemostatic Adjunct	Use/Dosing Guidelines	Article
rFVIIa	20–180 µg/kg, if >3 TBV loss (Dehmer) 50–90 µg/kg, maximum 3 doses (Dressler) Physician discretion beyond MTP (Chidester) Tailor to coagulation results (Craig) Single dose available beyond MTP (Hendrickson) 24% of sites use (Horst)	Horst 2016 Navaratnam 2016 Chidester 2012 Craig 2012 Hendrickson 2012 Dehmer 2010 Dressler 2010
Antifibrinolytics (tranexamic acid [TXA], aminocaproic acid)	1 g TXA IV within 3 h of injury and re-dose based on assessment* (Eckert) 10 mg/kg (max 1 g) TXA IV over 10 min if fibrinogen <1 g/L (Kua) 15% of sites use (Horst)	Horst 2016 Neff 2015 Eckert 2014 Kua 2014
Fibrinogen concentrate	Tailor to coagulation results (Craig) 13% of sites use (Horst)	Horst 2016 Craig 2012
Prothrombin complex concentrate	Tailor to coagulation results (Craig) 11% of sites use (Horst)	Horst 2016 Navaratnam 2016 Craig 2012
Cryoprecipitate	>30 kg: MT packs no. 2–4 include 5 U (Hwu) 5 mL/kg over 10 min if fibrinogen <1 g/L (Kua) Alternate with PLT, weight-based (Diab) 0–4 kg: 2 U 5–9 kg: 3 U 10–24 kg: 4 U 25–49 kg: 6 U ≥50 kg: 8 U Weight-based (Craig) <5 kg: 1 U 5–10 kg: 2 U 10–20 kg: 1 U (pooled) >20 kg: 2 U (pooled) Alternate with PLT, weight-based (Hendrickson) 1–5 kg: 1 U 6–10 kg: 2 U 11–25 kg: 4 U 26–50 kg: 6 U >50 kg: 8 U 0.1 U/kg (Dressler) 4 mL/kg if fibrinogen <1.0–1.5 g/L, or ongoing bleeding after administration of 1 pack RBC, FFP, PLT (Dehmer)	Hwu 2016 Navaratnam 2016 Livingston 2014 Kua 2014 Diab 2013 Craig 2012 Hendrickson 2012 Dressler 2010 Dehmer 2010

*Adult dosing parameters.

MTP, massive transfusion protocol; TBV, total blood volume; PLT, platelets; RBC, red blood cells.

cryoprecipitate added to protocol packages.^{11,13,18} Cryoprecipitate transfusion was recommended based on a fibrinogen threshold <1.0–1.5 g/L.¹² Use of rFVIIa was variable, with some suggesting 50–90 µg/kg at a maximum three doses, whereas others report using the product at doses of 20–180 µg/kg only if a patient has lost greater than three TBV.^{12,15} TXA also demonstrated mixed dosing regimens, with some reports of adult boluses of 1.0 g within 3 h of injury whereas others used laboratory-based thresholds to dose 10 mg/kg (maximum 1 g).^{16,23}

Outcomes Comparing Massively Transfused Children With Those Receiving Non-Protocolized Transfusions

Outcomes were reviewed in two fashions in this review, the first being a descriptive comparison of children receiving MT with those who received sub-MT or no blood products. Not surprisingly, all five studies reporting mortality outcomes based on extent of transfusion found those who received MT had statistically significant higher mortality rates ranging from

5.4% to 50.6% as compared with 2.1–20.0%. Similar trends existed for other outcomes including total and ICU lengths of stay, ventilator days, and rates of bleeding or thrombosis.

Outcomes After Implementation of Massive Transfusion Protocols

The second outcome comparison assessed differences pre-MTP and post-MTP implementation by matching controls before and after MTP implementation (Table 5). In-hospital mortality was not significantly different after the implementation of a MTP in any study, with pre-MTP mortality rates reported at 23.0–53.8% and post-MTP rates of 35.3–47.1%. Similarly, ICU and total lengths of stay, and ventilator use did not change. Chidester et al. reported a significant decrease in bleeding and thrombosis rate after initiating a MTP from 12.0% to 0.0% ($p = 0.04$).⁹ Hwu et al. reported no change in time to administration of first blood product ($p = 0.68$), but did report shorter times to first FFP administration from 2.7 to 1.0 hours ($p = 0.005$) after MTP implementation.²¹

TABLE 5. Comparative Outcomes Between Patients Treated Before MTP vs. After MTP Implementation

Outcome	Article	Post-MTP Implementation	Before MTP Implementation	<i>p</i>
In-hospital mortality	Hwu et al. 2016	47.1%	53.8%	0.729
	Hwu et al. 2016†	35.3%	38.5%	0.994
	Chidester et al. 2012	45%	45%	>0.05
	Hendrickson et al. 2012	38%	23%	0.10
Total LOS (days) (mean)	Hwu et al. 2016	45.8	39.0	0.619
ICU LOS (days) (mean)	Hwu et al. 2016‡	6.0	4.3	0.330
	Hendrickson et al. 2012*	7.0	9.0	0.54
Ventilator use (days)	Hwu et al. 2016§	8.3	7.0	0.584
	Hendrickson et al. 2012	2.0	6.0	0.27
Bleeding/thrombosis	Chidester et al. 2012	0%	12%	0.04
Hours to first blood product (mean)	Hwu et al. 2016	0.9	0.8	0.688
Hours to first RBC (mean)	Hwu et al. 2016	1.4	0.8	0.180
Hours to first FFP (mean)	Hwu et al. 2016	1.0	2.7	0.005
	Hendrickson et al. 2012	0.8	3.3	<0.001
Hours to first PLT (mean)	Hwu et al. 2016	4.4	6.0	0.421

*Median value reported.

†24-h mortality.

‡ICU-free days reported.

§Ventilator-free days reported.

MTP, massive transfusion protocol; LOS, length of stay; RBC, red blood cells; FFP, fresh frozen plasma; PLT, platelets.

Local Experience With Pediatric Massive Transfusion

In total, 14 MTP activations occurred in the 2-year study period at our institution. Of these, the most common indications were blunt (42.9%) and penetrating trauma (42.9%). The median patient age was 16.0 years (SD 5.4 years), with one patient ≤1 year old. The emergency department was the most common activation location (50%), with a median time of 11 min from arrival to activation. The in-hospital mortality of MTP patients was 50%, with all deaths occurring within 24 h of activation. Median blood product volume issued was 33.5 U (SD 20.9), and median volume transfused was 12.5 U (SD 18.4). Assuming a standard volume of 300 mL per RBC unit, the median volume of transfusion was 38.1 mL/kg (SD 40.3 mL/kg). Nearly 36.0% (5/14) of patients were administered a hemostatic adjunct (all tranexamic acid) and 50.0% (7/14) were monitored with viscoelastic tests (rotational thromboelastometry) for coagulopathy at physician discretion. The median ratio of RBC/FFP transfused was 1.5:1.

DISCUSSION

Children requiring massive blood product transfusion have a high mortality, ranging from 33% to 50% in most contemporary series outside of the military experience. In fact, their mortality and morbidity were substantially higher than those who require sub-MT or no transfusion, making this group of patients a critical group to focus research efforts in both pediatric critical care and trauma. Improving the survival of these children through further research and adoption of best practices should be a priority. In our study, the most comprehensive systematic review of the contemporary experience with pediatric MT to date, we found no prospective studies to guide management of these children and very few large studies. Furthermore, we found a variety of definitions, protocols, and practices related to pediatric

MT. Although the current literature is not robust enough to highlight best practices, we identified the following critical areas for further discussion and study:

- Need for a common working definition of pediatric MT for consistency in clinical guidelines and research
- Further study of effectiveness of MTP in improving mortality
- Need for more research into initial triggers for MT in children and goal-directed therapy endpoints
- National-level studies of whether balanced transfusion ratios affect mortality in pediatric MT
- Understanding best practices regarding hemostatic adjuncts in specific groups of children requiring MT (i.e., trauma patients, cardiac surgery patients, etc.)

We found that only half of the studies we reviewed reported having formal MT guidelines. Unfortunately, in aggregating these studies, we found that implementation of an MTP did not significantly reduce mortality or major morbidity.^{9,18,22} This finding runs counter to the adult experience where the literature so clearly demonstrates reduction of mortality and blood product use in adult trauma patients after adoption of an MTP that the American College of Surgeons advocated for MTP development in adult trauma centers.²⁴ This paradox may be explained by the lack of adherence to protocol guidelines for blood product ratios in the pediatric studies reviewed, which could have mitigated expected mortality benefits. We did find that the pediatric MTPs did facilitate product availability, as shown in two different studies that time to FFP is shorter after MTP implementation.^{18,22}

We also noted the wide range of definitions for MT in children across studies. Although all definitions constitute high blood loss, some are more extreme (Schroeder et al. >50% TBV in 1 h) than others (Hwu et al. >50% TBV within 24 h).^{21,22,29} Lack of a consistent and widely used definition may hamper developments of guidelines that can be used across

institutions or in research. However, the variety in definitions for pediatric MT may reflect a shift in definition over time. The article published by Neff et al. in 2015 provided one of the largest cohorts of pediatric MTP patients and assessed this definition of MT, finding that the 40 mL/kg total blood product threshold most reliably identified pediatric patients at risk for early or in-hospital mortality.²⁷ This definition has since been cited in more recent literature, and we would encourage its adoption as a standard definition of pediatric MT.

Adult literature supports balanced 1:1:1 ratio transfusion of RBC/FFP/PLT (platelets) in the initial approach to MT; however, this emphasis on a balanced ratio of blood products has not been fully adopted in pediatric MT.² Unfortunately, there is little direct evidence to support balanced ratio transfusion in pediatric patients. Often, varying blood product ratios are described based on weight thresholds of children. This diversity in ratios may try to account for age-related changes in hemostasis and thrombosis. However, these weight thresholds for defining ratios are ill-defined and appear arbitrary, as many centers use different weight thresholds. Another downside to weight-based approaches with multiple thresholds is the additional complexity integrated into the protocol that may introduce delays or logistic complications to release and delivery of blood products. Finally, a pediatric patient's weight might be significantly affected by volume loss in the form of previous or ongoing hemorrhage, leading to underestimation of transfusion needs or miscategorizing into other weight categories.

All of the studies included in our analysis including more than one patient demonstrated a discrepancy in the ratio of blood products prescribed in their protocol versus what was actually administered to the patients, regardless if a 1:1:1 ratio or other was targeted. The ratios always differed from the prescribed protocol in favor of additional RBC transfusions. Potential explanations for this discrepancy include longer thawing times for plasma which may delay a full round of transfusion products from being administered before the next round of RBCs reaching the patient. A second explanation for this observation is that physician discretion could ultimately affect the transfusion pattern despite institutional MT guidelines. This could be in part caused by lack of clinician education about the targeted ratios of blood products, especially at institutions that have specified blood bank-specific procedures. If clinicians are unaware of the target ratio, their product administration may favor packed red cells. A third explanation is that emergency release products may contain red blood cells only and may therefore skew the results despite subsequent rounds of balanced transfusion. Interestingly, at institutions that administer both RBCs and FFP in their emergency release products, the overall ratios administered are closer to target ratios than at institutions administering only RBC in the emergency release, which supports this conjecture. This is demonstrated by Hendrickson et al. who targeted a 1:1:0.5 ratio in all age groups.¹⁸ Their administered ratio of RBC/FFP was 1.8:1, which was much closer to target than their 6.7:1 RBC/PLT ratio. Because platelets are not administered in emergency release, the ratio was skewed heavily in favor of RBC. Finally, some advanced pre-hospital practitioners may have packed red cells available for transfusion in the field before arrival at an institution with a pediatric MTP in place, similarly skewing the overall transfusion ratios toward RBCs.

The lack of consensus on an appropriate trigger for MTP activation mirrors the inconsistencies in definition for pediatric MT. Physician discretion, although the most common trigger for activation, remains highly subjective, especially when paired with various volume thresholds as decision-making tools. This highlights areas for future research in predicting which pediatric patients will require large volume transfusions. Although 78% of protocols identified in this review included initial and repeat laboratories to monitor the status of transfusion, less than a third defined target values for goal-directed therapy. Studies have assessed laboratory values indicating greater odds for receipt of MT, but none highlight outcomes based on monitored laboratory values.^{9,21} This could explain the lack of clear guidelines for target laboratory values as a guide to goal-directed transfusion therapy. A lack of target values may also highlight the difficulty in stabilizing components like platelets and fibrinogen, which are subject to significant changes during large volume transfusion. However, with a lack of target laboratory values or goal-directed measures, it is difficult to standardize treatments across patients and clearly define timing of patient stability or the cessation of protocol efforts. Pediatric patients may remain under-resuscitated or conversely receive excess blood products without targeted laboratory values as an adjunct to clinician discretion determination of resuscitation end-points. Notable laboratories that remain underused in these protocols include ROTEM and TEG. These two global assessments of hemostatic function have been shown to decrease blood product use when implemented as part of transfusion protocols.^{33–35} Similar to our findings of underuse, Horst et al. found that only 9% of U.S. centers used TEG in monitoring resuscitative efforts.¹⁹

A survey of U.S. and Canadian pediatric MTPs demonstrated that the majority of sites do not or incorporate hemostatic adjuncts in their protocols.¹⁹ This included a 24% prevalence of the use of rFVIIa, 15% prevalence of antifibrinolytic use, 13% use of fibrinogen concentrate, and 11% use of prothrombin complex concentrate. Often, their use is not formally delineated in the MTP, but instead are available at physician discretion with limited guidelines for use.^{9,18} Antifibrinolytics, most often TXA, were not routinely cited in the available protocols. Recent evidence from adult trauma literature in the CRASH-2 trial suggests that early administration of TXA may reduce mortality while also conferring a cost-effective approach to MT.³⁶ Cryoprecipitate similarly did not appear in many MTPs, or was only used in the late packages released. This seems to be an opportunity for improvement in many of these protocols, as low fibrinogen has been independently associated with higher rates of mortality in adult trauma patients, and earlier administration of fibrinogen has been suggested as a method of improving outcomes.³⁷ Further outcomes studies in pediatric transfusion with the incorporation of early administered hemostatic adjuncts are necessitated to assess their benefit and proper implementation within pediatric MTPs.

Our local experience reflected the same themes we found across studies. Despite having a stated goal of balanced transfusion ratios, we found discrepancies between ratios of blood products transfused and target ratios even after implementation of our pediatric MTP. Hemostatic adjunct use was at the

discretion of the treating physician and highly variable. Our mortality, consistent with existing reports, remained high for these patients.

Limitations

This study is subject to limitations inherent to systematic reviews. First, because of the limited amount of published research in this topic, the final list of publications analyzed was relatively heterogeneous. Not only were the studies conducted in different formats but they were performed in a variety of settings, including major trauma centers, stand-alone children's hospitals, or in pediatric units of a general hospital. MT also occurred in a variety of patient populations, including pediatric patients presenting for trauma or high blood volume loss during cardiac and transplant surgery. Because of this heterogeneity, we were unable to perform a formal meta-analysis. A second limitation but important finding of this study is that all articles do not define MT similarly. This makes generalizing findings among studies difficult.

CONCLUSION

In summary, this systematic review provides highlights of current practice strategies in pediatric MT. Our institutional experience is consistent with the broader national and international experience in regards to mortality and protocol adherence. Centers hold the potential to improve with respect to protocol adherence and systematic use of hemostatic adjuncts in this pediatric population. This review highlights the scattered, heterogeneous quality of studies in this field. Ultimately, prospective, multi-institutional studies would be helpful to more formally and systematically assess MTPs in this unique and diverse patient population to target optimal protocols and improve patient outcomes.

AUTHORSHIP

R.W.K. contributed to literature search, study design, data interpretation, writing, and critical revision. H.J.L. contributed to literature search, study design, data interpretation, and writing. C.R. contributed to literature search, data interpretation, and critical revision. C.M.R. contributed to data interpretation, writing, and critical revision. U.P.N. contributed to literature search, study design, data interpretation, and critical revision. J.L.P. contributed to literature search, primary data collection, data analysis, data interpretation, and writing. E.T.T. contributed to literature search, study design, data interpretation, writing, and critical revision.

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REFERENCES

- Bhangu A, Nepogodiev D, Doughty H, Bowley DM. Meta-analysis of plasma to red blood cell ratios and mortality in massive blood transfusions for trauma. *Injury*. 2013;44(12):1693–1699.
- Holcomb JB, Tilley BC, Baraniuk S, Fox EE, Wade CE, Podbielski JM, Junco DJ, Brasel KJ, Bulger EM, Calicut AL, et al. Transfusion of plasma, platelets, and red blood cells in a 1:1:1 vs a 1:1:2 ratio and mortality in patients with severe trauma: the PROPPR randomized clinical trial. *JAMA*. 2015;313(5):471–482.
- Jaffray J, Young G. Developmental hemostasis: clinical implications from the fetus to the adolescent. *Pediatr Clin North Am*. 2013;60(6):1407–1417.
- Acker SN, Hall B, Hill L, Partrick DA, Bensard DD. Adult-based massive transfusion protocol activation criteria do not work in children. *Eur J Pediatr Surg*. 2017;27(1):32–35.
- Agrawal A, Beethe AB, Sullivan JN, Jones BM, Adams JJ, Duhacheck-Stapleman AL. Continuous hemoglobin monitoring during massive blood transfusion in a multivisceral pediatric transplant patient. *J Clin Anesth*. 2013;25(7):578–581.
- Arul GS, Reynolds J, DiRusso S, Scott A, Bree S, Templeton P, Midwinter MJ. Paediatric admissions to the British military hospital at Camp Bastion, Afghanistan. *Ann R Coll Surg Engl*. 2012;94(1):52–57.
- Brown KA, Bissonnette B, MacDonald M, Poon AO. Hyperkalaemia during massive blood transfusion in paediatric craniofacial surgery. *Can J Anaesth*. 1990;37(4 Pt 1):401–408.
- Buntain SG, Pabari M. Massive transfusion and hyperkalaemic cardiac arrest in craniofacial surgery in a child. *Anaesth Intensive Care*. 1999;27(5):530–533.
- Chidester SJ, Williams N, Wang W, Groner JL. A pediatric massive transfusion protocol. *J Trauma Acute Care Surg*. 2012;73(5):1273–1277.
- Cote CJ, Liu LM, Szyfelbein SK, Goudsouzian NG, Daniels AL. Changes in serial platelet counts following massive blood transfusion in pediatric patients. *Anesthesiology*. 1985;62(2):197–201.
- Craig RG, Diwan R, Kuchi S. The management of massive hemorrhage in children: Development of a local and regional toolkit. *Paediatr Anaesth*. 2012;22(9):910–911.
- Dehmer JJ, Adamson WT. Massive transfusion and blood product use in the pediatric trauma patient. *Semin Pediatr Surg*. 2010;19(4):286–291.
- Diab YA, Wong EC, Luban NL. Massive transfusion in children and neonates. *Br J Haematol*. 2013;161(1):15–26.
- Downes K, Sarode R. Massive blood transfusion. *Indian J Pediatr*. 2001;68(2):145–149.
- Dressler AM, Finck CM, Carroll CL, Bonanni CC, Spinella PC. Use of a massive transfusion protocol with hemostatic resuscitation for severe intraoperative bleeding in a child. *J Pediatr Surg*. 2010;45(7):1530–1533.
- Eckert MJ, Wertin TM, Tyner SD, Nelson DW, Izenberg S, Martin MJ. Tranexamic acid administration to pediatric trauma patients in a combat setting: the pediatric trauma and tranexamic acid study (PED-TRAX). *J Trauma Acute Care Surg*. 2014;77(6):852–858; discussion 8.
- Edwards MJ, Lustik MB, Clark ME, Creamer KM, Tuggle D. The effects of balanced blood component resuscitation and crystalloid administration in pediatric trauma patients requiring transfusion in Afghanistan and Iraq 2002 to 2012. *J Trauma Acute Care Surg*. 2015;78(2):330–335.
- Hendrickson JE, Shaz BH, Pereira G, Parker PM, Jessup P, Atwell F, Polstra B, Atkins E, Johnson KK, Bao G, et al. Implementation of a pediatric trauma massive transfusion protocol: one institution's experience. *Transfusion*. 2012;52(6):1228–1236.
- Horst J, Leonard JC, Vogel A, Jacobs R, Spinella PC. A survey of US and Canadian hospitals' paediatric massive transfusion protocol policies. *Transfus Med*. 2016;26(1):49–56.
- Huang CJ, Cheng KW, Chen CL, Wu SC, Shih TH, Yang SC, Jawan B, Wang CH. Predictive factors for pediatric patients requiring massive blood transfusion during living donor liver transplantation. *Ann Transplant*. 2013;18:443–447.
- Hwu RS, Keller MS, Spinella PC, Baker D, Shi J, Leonard JC. Identifying potential predictive indicators of massive transfusion in pediatric trauma. *Acad Emerg Med*. 2015;22(5):S33–S34.
- Hwu RS, Spinella PC, Keller MS, Baker D, Wallendorf M, Leonard JC. The effect of massive transfusion protocol implementation on pediatric trauma care. *Transfusion*. 2016.
- Kua JP, Ong GY, Ng KC. Physiologically-guided balanced resuscitation: an evidence-based approach for acute fluid management in paediatric major trauma. *Ann Acad Med Singapore*. 2014;43(12):595–604.
- Lee AC, Reduque LL, Luban NL, Ness PM, Anton B, Heitmiller ES. Transfusion-associated hyperkalemic cardiac arrest in pediatric patients receiving massive transfusion. *Transfusion*. 2014;54(1):244–254.

25. Livingston MH, Singh S, Merritt NH. Massive transfusion in paediatric and adolescent trauma patients: incidence, patient profile, and outcomes prior to a massive transfusion protocol. *Injury*. 2014;45(9):1301–1306.
26. Navaratnam M, Ng A, Williams GD, Maeda K, Mendoza JM, Concepcion W, Hollander SA, Ramamoorthy C. Perioperative management of pediatric en-bloc combined heart–liver transplants: a case series review. *Paediatr Anaesth*. 2016;26(10):976–986.
27. Neff LP, Cannon JW, Morrison JJ, Edwards MJ, Spinella PC, Borgman MA. Clearly defining pediatric massive transfusion: cutting through the fog and friction with combat data. *J Trauma Acute Care Surg*. 2015;78(1):22–28; discussion 28–9.
28. Nosanov L, Inaba K, Okoye O, Resnick S, Upperman J, Shulman I, Rhee P, Demetriades D. The impact of blood product ratios in massively transfused pediatric trauma patients. *Am J Surg*. 2013;206(5):655–660.
29. Schroeder HG, Forbes AR. Massive blood replacement in neonates and children. *Br J Anaesth*. 1969;41(11):953–961.
30. Shroyer MC, Griffin RL, Mortellaro VE, Russell RT. Massive transfusion in pediatric trauma: analysis of the National Trauma Databank. *J Surg Res*. 2017;208:166–172.
31. Smith SA, Livingston MH, Merritt NH. Early coagulopathy and metabolic acidosis predict transfusion of packed red blood cells in pediatric trauma patients. *J Pediatr Surg*. 2016;51(5):848–852.
32. Sparkle T, Cameron S. Massive transfusion of 5 U packed red blood cells, 3 U fresh frozen plasma, and 160 cc of platelets in a 14-month-old patient. *Am J Case Rep*. 2016;17:211–213.
33. Gorlinger K, Dirkmann D, Hanke AA, Kamler M, Kottenberg E, Thielmann M, Jakob H, Peters J. First-line therapy with coagulation factor concentrates combined with point-of-care coagulation testing is associated with decreased allogeneic blood transfusion in cardiovascular surgery: a retrospective, single-center cohort study. *Anesthesiology*. 2011;115(6):1179–1191.
34. Schaden E, Kimberger O, Kraincuk P, Baron DM, Metnitz PG, Kozek-Langenecker S. Perioperative treatment algorithm for bleeding burn patients reduces allogeneic blood product requirements. *Br J Anaesth*. 2012;109(3):376–381.
35. Whiting P, Al M, Westwood M, Ramos IC, Ryder S, Armstrong N, Misso K, Ross J, Severens J, Kleijnen J. Viscoelastic point-of-care testing to assist with the diagnosis, management and monitoring of haemostasis: a systematic review and cost-effectiveness analysis. *Health Technol Assess*. 2015;19(58):1–228, v-vi.
36. Roberts I, Shakur H, Coats T, Hunt B, Balogun E, Barnetson L, Cook L, Kawahara T, Perel P, Prieto-Merino D, et al. The CRASH-2 trial: a randomised controlled trial and economic evaluation of the effects of tranexamic acid on death, vascular occlusive events and transfusion requirement in bleeding trauma patients. *Health Technol Assess*. 2013;17(10):1–79.
37. Rourke C, Curry N, Khan S, Taylor R, Raza I, Davenport R, Stanworth S, Brohi K. Fibrinogen levels during trauma hemorrhage, response to replacement therapy, and association with patient outcomes. *J Thromb Haemost*. 2012;10(7):1342–1351.